第4次作业 4.A 3.1 3.5 3.6 3.15 3.17 4.B 3.19 3.21 3.25 3.29

3.1 $\int_0^t \beta dt = \int_0^\omega d\omega \Rightarrow \omega = \beta t \Rightarrow \omega = 20 rad/s$

$v = \omega R \Rightarrow v = 4 m/s$

$\begin{cases} a_n = \frac{v^2}{R} \\ a_\tau = \beta R \Rightarrow a = 30.02 m/s^2 \\ a = \sqrt{a_n^2 + a_\tau^2} \end{cases}$

$\int_0^t \omega dt = \int_0^\theta d\theta \Rightarrow \theta = \frac{1}{2}\beta t^2 \Rightarrow \theta = 20 rad$

$n = \frac{\theta}{2\pi} \Rightarrow n = 3.18$

3.5 (1) $J_0 = 3mR^2 + 2mR^2 + 2mR^2 + mR^2 \quad J_0 = 8mR^2$

$X = \frac{3mR - mR}{8m} \Rightarrow X = \frac{1}{4}R$

(2) $J_1 = 3m(R-X)^2 + m(R+X)^2 + 4m(X^2+R^2) \quad J_1 = \frac{11}{2}mR^2$

3.6 $dJ = x^2 dm = x^2 \rho_l dx = Kx^3 dx$

$J = \int_0^L Kx^3 dx = \frac{1}{4}Kx^4 \Big|_0^L = \frac{1}{4}KL^4$

3.15 $\begin{cases} mgR = J\beta + maR \\ \beta R = a \\ mg - T = ma \\ J = \frac{1}{2}MR^2 \end{cases} \Rightarrow \begin{cases} \beta = \frac{2mg}{MR+2mR} \\ a = \frac{2mg}{M+2m} \\ T = \frac{mMg}{M+2m} \end{cases}$

$\Rightarrow \begin{cases} \beta = 0.495 rad/s^2 \\ a = 0.099 m/s \\ T = 0.99 N \end{cases}$

3.17 $\begin{cases} m_1 gr - Mf - \mu m_2 gr = J\beta + (m_1+m_2)ar \\ \beta r = a \\ m_1 g - T_1 = m_1 a \\ T_2 - \mu m_2 g = m_2 a \end{cases} \Rightarrow \begin{cases} a = \frac{m_1 g - \frac{Mf}{r} - \mu m_2 g}{\frac{J}{r^2} + m_1 + m_2} \\ T_1 = \frac{mg\frac{J}{r^2} + m_1 m_2 g + \frac{m_1 Mf}{r} + \mu m_1 m_2 g + \mu m_2 g \frac{J}{r^2}}{\frac{J}{r^2} + m_1 + m_2} \\ T_2 = \frac{\mu m_1 m_2 g + m_1 m_2 g + \mu m_2 g \frac{J}{r^2} - m_2 \frac{Mf}{r}}{\frac{J}{r^2} + m_1 + m_2} \end{cases}$

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3.19

$$\begin{cases} mg-2T=ma \\ TR=J\beta \\ a=\frac{1}{2}\beta R \\ J=\frac{1}{2}MR^2 \end{cases} \Rightarrow a=\frac{mg}{2M+m}$$

$a=3m/s^2$

3.21

$$\begin{cases} J_1=\frac{2}{5}MR^2 \\ mgh=\frac{1}{2}mv^2+\frac{1}{2}J\omega_0^2+\frac{1}{2}J_1\omega_1^2 \\ \omega_0 r=\omega_1 R=V \end{cases} \Rightarrow V=\sqrt{\frac{10mghr^2}{5mr^2+5J+2MR^2}}$$

3.25

$$\frac{3}{2}mgl=\frac{1}{2}J\omega^2+\frac{1}{2}mv^2$$

$$\begin{cases} v=\omega l,\ a=\beta l \\ J=\frac{1}{3}ml^2 \\ \frac{1}{2}mgl+mgl=J\beta+mal \\ F_x=2m\omega^2\frac{3}{4}l \\ \Rightarrow mg-F_y=2m\beta\frac{3}{4}l \end{cases} \Rightarrow \begin{cases} \omega=\frac{3}{2}\sqrt{\frac{g}{l}} \\ \beta=\frac{9g}{8l} \\ \vec{F}=\frac{27}{8}mg\vec{i}+\frac{25}{16}mg\vec{j} \end{cases}$$

3.29

$$\begin{cases} mgl\sin\theta=\frac{1}{2}Kl^2 \\ mgx\sin\theta=\frac{1}{2}mv^2+\frac{1}{2}J\omega^2+\frac{1}{2}Kx^2 \\ v=\omega r \end{cases} \Rightarrow \begin{cases} l=\frac{2mg\sin\theta}{K} \\ v=\sqrt{\frac{2mgx\sin\theta+Kx^2}{m^2+\frac{J}{r^2}}} \end{cases}$$

∴当 $x=\frac{mg\sin\theta}{K}$ 时，v 有最大值

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